

⑤ Complex Numbers

- De Moivre's Theorem: $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$
- Proof of de Moivre's theorem: use induction
- Use of dMT:

1) Trig ratios of fundamental

angle: $z = C + iS$,

$$z^n = (C + iS)^n$$

Use: $\operatorname{Re}(z^n) = \cos(n\theta)$, $\rightarrow$ also for $\sec(n\theta)$

$\operatorname{Im}(z^n) = \sin(n\theta)$ $\rightarrow$ also for $\operatorname{cosec}(n\theta)$

2) Powers of $\sin \theta$ and

$\cos \theta$ in terms of other

angles: Use $z + \frac{1}{z} = e^{i\theta} + e^{-i\theta} = 2\cos \theta$

$$z - \frac{1}{z} = e^{i\theta} - e^{-i\theta} = 2i\sin \theta$$

and raise them to different powers.

3) n^{th} roots of unity: remember to find all basic angles first, and then use the reduction through dMT. $r = \sqrt{(C)^2 + (S)^2}$, $\theta = \tan^{-1}\left(\frac{S}{C}\right)$

4) Series: $e^{i\theta} + e^{-i\theta} = 2\cos \theta$, $\rightarrow z + 1/z$

$$e^{i\theta} - e^{-i\theta} = 2i\sin \theta \rightarrow z - 1/z$$

Always conjugate denominator.

Multiply by $\frac{\bar{i}}{i}$ to remove i in denominator.